

TRAINING SCENARIO — NOT AN ACTUAL TASKING

NETWORK INSTALLATION

TASK ORDER

Building 1042 · Administrative Facility

Facility	Building 1042 — two floors, newly renovated, structurally cabled
Tasking	Install, configure, secure, and turn over the building network
Issued to	You. Your SOP is issued individually and is not transferable.
Work packages	Four. Each is signed off by your instructor before you continue.
Turn-in	Completion report generated by the system at the end of Work Package 4

1. Situation

Building 1042 has just come out of renovation. The contractor pulled cable and terminated the drops, the wall mounts are in place in each work area, and the carrier has landed a circuit on the demarcation point in the IT room. Everything above that is bare.

Both communications closets are empty racks. The equipment for the job is staged in a box in the IT room. Nothing is configured. No device has a hostname, no VLAN exists, no port is assigned, and there is no route off the building.

You are the technician assigned to stand it up.

1.1 Your SOP

You have been issued a Standard Operating Procedure for this installation. It contains the addressing plan, VLAN assignments, cable run schedule, circuit handoff letter, wireless parameters, and the credential standard for the facility. Press Tab at any time to open it. It stays open while you work.

Your SOP is issued to you specifically. The technician beside you is working a different set of numbers. Configure to your SOP — copying a neighbor's screen produces a wrong answer.

If a value is not in this task order, it is in your SOP. If it is in neither, you are expected to derive it from what you have been taught.

1.2 Standards

This is a turnover, not a lab exercise. The following are held to standard throughout and are not called out again in each work package:

- Device names match the SOP exactly, including case and punctuation.
- Every managed device receives the full hardening baseline before it carries traffic.
- Ports that are not in service are secured, not left in their default state.
- Cable type is selected for the connection, not for convenience.
- Configuration is saved. Unsaved configuration does not survive a power event.
- Connectivity is verified from the user endpoint, not from the device you just configured.

2. Mission

Install and configure the Building 1042 network across four work packages, and turn over a network that is segmented, routed, secured, connected to the carrier circuit, and serving guest wireless.

WP1 — Physical Installation

Populate both racks, place user equipment, and complete the cable plant.

DELIVERABLES

- Both racks populated per the SOP rack elevation, every device named.
- User equipment placed at its assigned work area.
- Inter-closet link established between the two switches with the correct cable type.
- Every user drop terminated through the structured path: workstation to wall mount, punch-down to patch panel, patch cord to switch.
- Router uplinked to the ground floor switch and the carrier circuit landed on the demarc.

ACCEPTANCE

Physical only. Nothing at this layer reports success or failure — a miswired run stays silent until you verify connectivity in WP3. Build it right the first time or plan on tracing it later.

WP2 — Layer 2

Harden both switches, build the VLAN database, and assign the switched infrastructure.

DELIVERABLES

- Both switches hardened to the baseline in your SOP.
- Full VLAN database built on each switch, named per the SOP.
- Access ports assigned to the VLAN matching the function of the device on them.
- Trunks established where the SOP calls for them, carrying the user VLANs with the correct native VLAN.
- Unused ports parked in the dead VLAN and shut down.
- Both configurations saved.

ACCEPTANCE

Your VLAN and trunk output on both switches reflects the SOP. Verify it yourself before you call the package complete.

► *A trunk carries tagged frames. It does not build anything on the far side for you.*

WP3 — Layer 3

Harden the router, provide inter-VLAN routing, and bring up the carrier circuit.

DELIVERABLES

- Router hardened to the same baseline as the switches.
- A gateway for each user VLAN, delivered over the single trunked uplink using 802.1Q subinterfaces.
- WAN interface addressed from the circuit handoff letter and brought up.
- A default route pointing at the provider.
- End-to-end connectivity verified from workstations, and the routing table confirmed.
- Configuration saved.

ACCEPTANCE

From a workstation: you reach your own gateway, you reach a host in a different VLAN on the other floor, and you reach the provider side of your circuit. On the router, the routing table shows a connected route for each user subnet and a static default.

► *Subinterfaces ride on the physical interface. If the physical interface is down, everything riding on it is down.*

WP4 — Wireless

Install and secure the guest access point in the lounge and prove it reaches the internet.

DELIVERABLES

- Access point installed at the designated ceiling mount and named per the SOP.
- AP uplink run through the patch panel to the switch, and that switchport placed in service in the correct VLAN.
- AP secured per the wireless sheet in your SOP: administrative credentials, SSID, encryption, WPS, and channel.
- Guest wireless verified end to end from the test laptop.

ACCEPTANCE

The test laptop associates to your SSID, receives addressing, reaches the provider, and loads a page. That page load proves every layer beneath it.

- ▶ *The port your AP lands on was secured during WP2. Putting it into service is a deliberate act.*

3. Execution

3.1 Moving and working

WASD / click	Move. Click a spot on the floor to walk there.
E	Interact with whatever you are standing at — rack, desk, equipment box, demarc, ceiling mount.
Tab	Open and close your SOP. It does not block your view.
Stairs	Walk the ramp to change floors.
Console	Open a rack panel and select Console to get a terminal on that device.
Workstation	Stand at a desk and press E to reach a PC's command prompt and browser.

3.2 Cabling

Click a port to start a run, then click the far port to land it. Select the cable type before you pull. Punch-down blocks land only on other punch-down blocks — the exception is a patch panel rear onto an active device port. Cable jackets are color-coded so you can audit your own work at a glance: blue for straight-through, orange for crossover, violet for permanent punch-down runs.

3.3 The terminal

The console behaves like the IOS you have already used. Command abbreviation works. A question mark gives you context help. The up arrow recalls history. Incomplete and ambiguous commands are rejected the way real IOS rejects them.

Malformed addressing is rejected rather than silently accepted. A double dot, an octet above 255, or a non-contiguous mask returns an invalid input error. If you get one, read what you typed before you retype it.

► *If you set a credential that does not match your SOP, the device accepts it — but the turnover check will not pass until it matches. You will get a notice when it happens. Do not ignore it.*

3.4 When something does not work

A failed ping returns a trace line naming the layer that broke: a missing cable, a shut port, a wrong VLAN, a missing gateway, an unassociated client. That is your starting point. Work that layer.

The discipline is the same one you have been taught: verify from the endpoint, isolate to a layer, then fix one thing at a time and re-test. Re-entering commands you have already entered is not troubleshooting.

3.5 Sign-off and turn-in

At the end of each work package the job stops until your instructor initials it. Have your verification output on screen when you call them over — be ready to show what you built and explain why it is correct.

When the final package is signed, the system generates your completion report: time per work package, instructor initials, and a verification code. Submit it as directed.

4. Command Reference

Syntax reminders only. Which command belongs where, and which values go in it, comes from your SOP and your previous labs. Values shown in angle brackets are yours to supply.

Modes and navigation

Command	Purpose
enable	Enter privileged EXEC mode
configure terminal	Enter global configuration mode
exit	Drop back one mode
end	Return to privileged EXEC from any config mode
do <command>	Run an EXEC command without leaving config mode
?	Context help — what is valid at this point
<partial>?	List commands beginning with those characters

Device hardening

Command	Purpose
hostname <name>	Set the device name
enable secret <password>	Set the encrypted privileged-mode password
service password-encryption	Encrypt plaintext passwords in the configuration
no ip domain-lookup	Stop mistyped commands resolving as hostnames
banner motd #<text>#	Set the login banner
line console 0	Enter console line configuration
line vty 0 4	Enter virtual terminal line configuration
password <password>	Set the password on the selected line
login	Require that password at login

VLANs

Command	Purpose
vlan <id>	Create a VLAN and enter its configuration
name <name>	Name the VLAN
state suspend	Suspend the VLAN — traffic in it is not forwarded
show vlan brief	List VLANs and their assigned ports

Switchports

Command	Purpose
interface <type><slot>/<port>	Select a single interface
interface range <if> - <n>	Select a range of interfaces
switchport mode access	Set the port as an access port
switchport access vlan <id>	Place the access port in a VLAN
switchport mode trunk	Set the port as a trunk

Command	Purpose
switchport trunk allowed vlan <list>	Restrict which VLANs cross the trunk
switchport trunk native vlan <id>	Set the untagged VLAN on the trunk
shutdown	Administratively disable the port
no shutdown	Bring the port up
show interfaces trunk	Verify trunk status, allowed list, and native VLAN

Routing

Command	Purpose
interface <if>	Select the physical interface
interface <if>.<subif>	Create or select a subinterface
encapsulation dot1q <vlan>	Tag the subinterface to a VLAN
ip address <address> <mask>	Assign an address
ip route <network> <mask> <next-hop>	Install a static route
ip route 0.0.0.0 0.0.0.0 <next-hop>	Install a default route
show ip route	Display the routing table
show ip interface brief	Summarize interfaces, addresses, and status

Verification and saving

Command	Purpose
show running-config	Display the active configuration
show startup-config	Display the saved configuration
copy running-config startup-config	Save the configuration
ping <address>	Test reachability

Workstation command prompt

Command	Purpose
ipconfig	Show the client address, mask, and gateway
ping <address>	Test reachability from the endpoint
tracert <address>	Show the path to a destination
cls	Clear the screen

5. Turnover Criteria

The installation is complete when all of the following are true. You should be able to demonstrate each on request.

- Every device carries its SOP name and the full hardening baseline.
- The VLAN database matches the SOP on both switches, and trunks agree on both ends.
- Ports not in service are parked and shut.
- A workstation reaches its own gateway, a host in another VLAN on the other floor, and the provider side of the circuit.
- The routing table shows a connected route for every user subnet and a default route to the provider.
- Guest wireless associates, is encrypted per the SOP, and reaches the internet.
- Every configuration is saved.

You built it. Be able to explain it. At each sign-off your instructor may ask why a value is what it is — "the SOP said so" is a starting point, not an answer.